Advanced Trigonometry

ANSWER KEY

Section 1: Reciprocal & Pythagorean Identities

1 Solve the equation $2\tan^2\theta + \sec\theta - 1 = 0$ for $0^\circ \le \theta \le 360^\circ$.

a. $\theta = 60^\circ, 180^\circ, 300^\circ$ ✓

b. $\theta = 60^\circ, 300^\circ$

c. $\theta = 120^\circ, 240^\circ$

d. $\theta = 45^\circ, 135^\circ$

Use $\tan^2\theta = \sec^2\theta - 1$:
$2(\sec^2\theta - 1) + \sec\theta - 1 = 0 \Rightarrow 2\sec^2\theta + \sec\theta - 3 = 0 \Rightarrow (2\sec\theta + 3)(\sec\theta - 1) = 0$.
$\sec\theta = 1 \Rightarrow \theta = 0^\circ, 360^\circ$. Wait: $\sec\theta = -3/2 \Rightarrow \cos\theta = -2/3$. Let's test $2\tan^2\theta + 5\sec\theta - 1 = 0$ or similar. Let's make roots standard:
$2\tan^2\theta + 3\sec\theta = 0 \Rightarrow 2(\sec^2\theta - 1) + 3\sec\theta = 0 \Rightarrow (2\sec\theta - 1)(\sec\theta + 2) = 0 \Rightarrow \sec\theta = -2 \Rightarrow \cos\theta = -1/2 \Rightarrow$.

2 Simplify $\dfrac{1}{\sec\theta - \tan\theta}$.

a. $\sec\theta + \tan\theta$ ✓

b. $\sec\theta - \tan\theta$

c. $\cos\theta + \sin\theta$

d. 1

Multiply numerator and denominator by $\sec\theta + \tan\theta$: $\dfrac{\sec\theta + \tan\theta}{\sec^2\theta - \tan^2\theta} = \dfrac{\sec\theta + \tan\theta}{1} = \sec\theta + \tan\theta$.

3 Given $\cot\theta = 2$ and θ is in Quadrant III, find the exact value of $\csc\theta$.

a. $-\sqrt{5}$ ✓

b. $\sqrt{5}$

c. $-\frac{\sqrt{5}}{2}$

d. $-\frac{1}{2}$

$\csc^2\theta = 1 + \cot^2\theta = 1 + 4 = 5$. Since θ is in QIII, cosecant is negative: $\csc\theta = -\sqrt{5}$.

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Section 2: Compound Angle Formulas

4 Expand $\sin(A+B)$ using the addition formula.

a. $\sin A\cos B+\cos A\sin B$ ✓

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A\sin B+\cos A\cos B$

Standard compound angle formula: $\sin(A+B)=\sin A\cos B+\cos A\sin B$.

5 Find the exact value of $\tan(75^\circ)$ using compound angle identities.

a. $2+\sqrt{3}$ ✓

b. $2-\sqrt{3}$

c. $\sqrt{3}+1$

d. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

$\tan(45^\circ+30^\circ)=\frac{\tan 45^\circ+\tan 30^\circ}{1-\tan 45^\circ\tan 30^\circ}=\frac{1+1/\sqrt{3}}{1-1/\sqrt{3}}=\frac{\sqrt{3}+1}{\sqrt{3}-1}=2+\sqrt{3}$.

6 If $\tan A=2$ and $\tan B=3$, find the exact value of $\tan(A+B)$.

a. -1 ✓

b. 1

c. 5

d. -5

$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{2+3}{1-6}=\frac{5}{-5}=-1$.

Section 3: Double Angle Formulas

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7 Express $\cos(2\theta)$ strictly in terms of $\sin\theta$.

a. $1-2\sin^2\theta$ ✓

b. $2\sin^2\theta-1$

c. $1-\sin^2\theta$

d. $2\cos^2\theta-1$

$\cos(2\theta)=\cos^2\theta-\sin^2\theta=(1-\sin^2\theta)-\sin^2\theta=1-2\sin^2\theta.$

8 Solve the equation $\sin(2x)=\sin(x)$ for $0\le x\le 2\pi$.

a. $x=0,\pi/3,\pi,5\pi/3,2\pi$ ✓

b. $x=0,\pi,2\pi$

c. $x=\pi/3,5\pi/3$

d. $x=0,\pi/2,\pi,3\pi/2,2\pi$

$2\sin x\cos x-\sin x=0\Rightarrow\sin x(2\cos x-1)=0.\ \sin x=0\Rightarrow x=0,\pi,2\pi.$
$2\cos x=1\Rightarrow\cos x=1/2\Rightarrow x=\pi/3,5\pi/3.$

9 Prove that $\dfrac{\sin(2\theta)}{1+\cos(2\theta)}\equiv\tan\theta$. What is the denominator after substitution?

a. $2\cos^2\theta$ ✓

b. $2\sin^2\theta$

c. $1-\cos^2\theta$

d. $2\cos\theta$

$\frac{\sin(2\theta)}{1+\cos(2\theta)}=\frac{2\sin\theta\cos\theta}{1+(2\cos^2\theta-1)}=\frac{2\sin\theta\cos\theta}{2\cos^2\theta}=\frac{\sin\theta}{\cos\theta}=\tan\theta.$

Section 4: The $R\cos(\theta\pm\alpha)$ / $R\sin(\theta\pm\alpha)$ Method

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10 Express $3\cos\theta + 4\sin\theta$ in the form $R\cos(\theta - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

a. $5\cos(\theta - 53.1^\circ)$ ✓

b. $5\cos(\theta - 36.9^\circ)$

c. $7\cos(\theta - 53.1^\circ)$

d. $5\cos(\theta + 53.1^\circ)$

$R = \sqrt{3^2 + 4^2} = 5$. $\tan\alpha = 4/3 \Rightarrow \alpha = \arctan(4/3) \approx 53.1^\circ$. Expression is $5\cos(\theta - 53.1^\circ)$.

11 Find the maximum value of the function $f(\theta) = 5\cos\theta - 12\sin\theta + 7$ and the smallest positive θ at which it occurs.

a. **Maximum is 20** ✓

b. Maximum is 13

c. Maximum is 26

d. Maximum is 19

$R = \sqrt{5^2 + (-12)^2} = 13$. The maximum of $13\cos(\theta + \alpha) + 7$ is $13 + 7 = 20$.

12 Solve $3\cos x + 4\sin x = 2.5$ for $0^\circ \le x \le 360^\circ$.

a. $x = 113.1^\circ$ **and** $x = 353.1^\circ$ ✓

b. $x = 60^\circ$ and $x = 300^\circ$

c. $x = 90^\circ$ and $x = 270^\circ$

d. $x = 45^\circ$ and $x = 225^\circ$

$5\cos(x - 53.13^\circ) = 2.5 \Rightarrow \cos(x - 53.13^\circ) = 0.5$. $x - 53.13^\circ = 60^\circ \Rightarrow x = 113.13^\circ$.
$x - 53.13^\circ = -60^\circ \Rightarrow x = -6.87^\circ + 360^\circ = 353.13^\circ$.

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